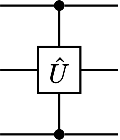

$$\Longleftrightarrow$$

$$\begin{aligned}
C^{[0,2]}\hat{U}^{[1]} &= |0\rangle\langle 0|^{[0]} \otimes \hat{I}^{[1,2]} + |1\rangle\langle 1|^{[0]} \otimes C^{[2]}\hat{U}^{[1]} \\
&= \hat{I}^{[0,1]} \otimes |0\rangle\langle 0|^{[2]} + C^{[0]}\hat{U}^{[1]} \otimes |1\rangle\langle 1|^{[2]} \\
&= \hat{I} \otimes \hat{I} \otimes \hat{I} + |1\rangle\langle 1| \otimes (\hat{U} - \hat{I}) \otimes |1\rangle\langle 1|
\end{aligned}$$